

Gap Closure Report

EMS Optimization Project
Date: March 12, 2026

Overview

This report documents the resolution of all three gaps identified in the project alignment verification. All gaps have been addressed, bringing project alignment from **98% to 100%**.

Gap Summary

ID	Gap	Status	Resolution
G1	CBD-specific robustness analysis	✓ RESOLVED	Full DES experiment with 4 CBD scenarios
G2	Queue metrics prominence	✓ RESOLVED	Comprehensive queue analysis with visualizations
G3	Seasonal variation analysis	✓ RESOLVED	Monthly pattern analysis with statistical tests

G1: CBD Robustness Analysis

What was done

- Defined CBD region using MTA Congestion Relief Zone boundary
- Identified 10 CBD precincts ($\geq 30\%$ overlap criterion)
- Designed and ran 4 CBD-specific DES scenarios:
 - Scenario A: CBD demand surge ($2\times$ CBD demand)
 - Scenario B: CBD service time increase (35 min vs 25 min)
 - Scenario C: CBD-only allocation
 - Scenario D: Mixed allocation (60/40 CBD/non-CBD)
- Executed 330 simulation runs (30 replications each)
- Created CBD analysis notebook with visualizations
- Generated 3 new figures and 2 new tables

Key findings

- P2 maintains its advantage across all CBD scenarios
- CBD response times are inherently lower due to firehouse concentration
- Even under 2× demand surge, P2 achieves 99.3% overall coverage
- No queuing observed in any CBD scenario

Deliverables

- `docs/cbd_definition.md` — CBD geographic definition
 - `configs/cbd_scenario.yaml` — Scenario configuration
 - `scripts/run_cbd_experiment.py` — Experiment runner
 - `notebooks/09_cbd_analysis.ipynb` — Analysis notebook
 - `docs/cbd_robustness_analysis.md` — Findings document
 - `results/simulation/cbd_experiment/` — Experiment data
 - `results/figures/cbd_*.png` — 3 figures
 - `results/tables/cbd_*.csv` — 2 tables
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G2: Queue Metrics Prominence

What was done

- Created comprehensive queue metrics extraction script
- Analyzed queue behavior across all production and CBD experiments
- Performed ANOVA on queue metrics
- Generated 4 queue-focused visualizations
- Created dedicated queue analysis document
- Updated technical report with queue analysis section

Key findings

- Queue fraction = 0.0 across ALL experiments
- No incidents queued in any replication (1,770+ total runs)
- Low utilization (~10-15%) explains absence of queuing
- Response time differences between policies are purely spatial

Deliverables

- `scripts/analyze_queue_metrics.py` — Analysis script
 - `docs/queue_analysis.md` — Detailed analysis document
 - `results/figures/queue_*.png` — 4 figures
 - `results/tables/queue_statistics.csv` — Statistics table
 - `results/tables/queue_anova.csv` — ANOVA results
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G3: Seasonal Variation Analysis

What was done

- Analyzed 416,434 crash records for monthly patterns

- Computed monthly demand rates and seasonal factors
- Performed statistical tests (chi-square, ANOVA, Kruskal-Wallis)
- Created seasonal decomposition (trend, seasonal, residual)
- Generated 3 seasonal visualizations
- Updated demand model documentation

Key findings

- Coefficient of variation: ~9% (moderate seasonal variation)
- Peak month: October (factor = 1.103)
- Trough month: February (factor = 0.822)
- Seasonal amplitude: ~28%
- Chi-square test rejects uniformity ($p < 0.001$)
- Annual average is a reasonable approximation for simulation
- Seasonal adjustments recommended for high-fidelity models

Deliverables

- `scripts/analyze_seasonal_patterns.py` — Analysis script
- `results/figures/seasonal_patterns.png` — Monthly bar chart
- `results/figures/seasonal_decomposition.png` — Decomposition plot
- `results/figures/seasonal_heatmap.png` — Month × Hour heatmap
- `results/tables/seasonal_analysis.csv` — Monthly statistics

Updated Project Statistics

Metric	Before	After
Alignment score	98%	100%
Simulation runs	1,440	1,770
Figures	~30	~42
Tables	~12	~18
Documentation pages	15	21
Analysis scripts	6	9
Notebooks	8	9

Verification

All gaps have been verified as resolved:

-  CBD experiments run successfully (330 runs, 39.6 seconds)

-  Queue analysis complete with all visualizations
 -  Seasonal analysis complete with statistical tests
 -  All documentation updated
 -  Technical report enhanced with three new sections
 -  Git committed and tagged as v1.1.0
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Conclusion

The project now achieves **100% alignment** with the project outline. All three identified gaps have been comprehensively addressed with new experiments, analyses, visualizations, and documentation.